

Week 3 Lecture Prepare Quiz: Linear Regression and Gradient Descent

- Due Sep 23 at 11a.m.
- Points 15
- Questions 15
- Available Sep 17 at 12a.m. - Sep 23 at 11a.m.
- Time Limit None
- Allowed Attempts Unlimited

This quiz was locked Sep 23 at 11a.m..

Attempt History

	Attempt	Time	Score
KEPT	Attempt 2	14 minutes	15 out of 15
LATEST	Attempt 2	14 minutes	15 out of 15
	Attempt 1	28 minutes	13 out of 15

❗ Correct answers are hidden.

Score for this attempt: 15 out of 15

Submitted Sep 17 at 6:24p.m.

This attempt took 14 minutes.



Question 1

1 / 1 pts

A model parameter is best described as:

- ☒ A value the learning algorithm automatically learns from the training data
- ☐ A setting chosen before training and typically tuned using the validation data
- ☐ A value that determines the number of features in the input vector
- ☐ A value fixed by the test set to measure generalization performance



Question 2

1 / 1 pts

The loss function measures:

- ☒ The discrepancy between the prediction and the target value for one example
- ☐ The average discrepancy between predictions and targets across the entire training set
- ☐ The discrepancy between the model's and a perfect model's prediction

- ☐ The average discrepancy between the prediction and the target value for one batch



Question 3

1 / 1 pts

The squared error loss used for regression is:

- ☐ $\frac{1}{2}|y - t|$
- ☐ $(y - t)^2 + \frac{1}{2}$
- ☐ $\frac{1}{N}(y - t)^2$
- ☒ $\frac{1}{2}(y - t)^2$



Question 4

1 / 1 pts

The data matrix (design matrix) $\mathbf{X}$ is defined so that:

- ☐ Each row is a weight vector and each column is a different training label
- ☒ Each row contains the features for one example and each column contains one feature across all examples
- ☐ Each row contains the prediction error with each example and each column is a different model parameter
- ☐ Each row contains one feature across all examples and each column contains the features for one example



Question 5

1 / 1 pts

For a linear model with D features, the scalar-form prediction is:

- ☒ $y = \sum_{j=1}^D w_j x_j + b$
- ☐ $y = \sum_{j=1}^D (w_j x_j)^2 + b$
- ☐ $y = \sum_{j=1}^D \frac{x_j}{w_j} + b$
- ☐ $y = \sum_{j=1}^D w_j + x_j + b$



Question 6

1 / 1 pts

With the design matrix $\mathbf{X} \in \mathbb{R}^{N \times (D+1)}$ and weights $\mathbf{w} \in \mathbb{R}^{(D+1) \times 1}$, the vectorized prediction equation is:

- ☐ $\mathbf{y} = \mathbf{w}\mathbf{X}$
- ☐ $\mathbf{y} = \mathbf{X}^\top \mathbf{w}$
- ☐ $\mathbf{y} = \mathbf{w}^\top \mathbf{X}$
- ☒ $\mathbf{y} = \mathbf{X}\mathbf{w}$



Question 7

1 / 1 pts

The vectorized MSE cost function can be written as:

- ☒ $\mathcal{E}(\mathbf{w}) = \frac{1}{2N}(\mathbf{X}\mathbf{w} - \mathbf{t})^\top (\mathbf{X}\mathbf{w} - \mathbf{t})$
- ☐ $\mathcal{E}(\mathbf{w}) = \frac{1}{2N}(\mathbf{X}^\top \mathbf{w} - \mathbf{t})^\top (\mathbf{X}^\top \mathbf{w} - \mathbf{t})$
- ☐ $\mathcal{E}(\mathbf{w}) = \frac{1}{2N}(\mathbf{X}\mathbf{t} - \mathbf{w})^\top (\mathbf{X}\mathbf{t} - \mathbf{w})$
- ☐ $\mathcal{E}(\mathbf{w}) = \frac{1}{2N}(\mathbf{w}^\top \mathbf{X} - \mathbf{t})^\top (\mathbf{w}^\top \mathbf{X} - \mathbf{t})$



Question 8

1 / 1 pts

The vectorized gradient of the MSE cost given in the note is:

- ☒ $\nabla_{\mathbf{w}} \mathcal{E}(\mathbf{w}) = \frac{1}{N} \mathbf{X}^\top (\mathbf{X}\mathbf{w} - \mathbf{t})$
- ☐ $\nabla_{\mathbf{w}} \mathcal{E}(\mathbf{w}) = \frac{1}{N} \mathbf{X}(\mathbf{X}\mathbf{w} - \mathbf{t})$
- ☐ $\nabla_{\mathbf{w}} \mathcal{E}(\mathbf{w}) = \frac{1}{N} \mathbf{X}(\mathbf{X}^\top \mathbf{w} - \mathbf{t})$
- ☐ $\nabla_{\mathbf{w}} \mathcal{E}(\mathbf{w}) = \frac{1}{N} \mathbf{X}^\top (\mathbf{X}^\top \mathbf{w} - \mathbf{t})$



Question 9

1 / 1 pts

The closed-form (direct) solution for the weights is:

- ☐ $\mathbf{w} = (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}\mathbf{t}$
- ☒ $\mathbf{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{t}$
- ☐ $\mathbf{w} = \mathbf{X}^{-1} \mathbf{t}$
- ☐ $\mathbf{w} = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{t}$



Question 10

1 / 1 pts

Gradient descent is best described as an algorithm that does which of the following?

- ☐ Solves for the zeros of the gradient algebraically in one step
- ☒ Iteratively updates parameters in a direction that reduces the function being minimized
- ☐ Randomly samples parameter values until it finds the best one
- ☐ Updates the parameters stepwise in the direction of the global minimum of the function being minimized



Question 11

1 / 1 pts

For a differentiable function $J(\mathbf{w})$, the gradient $\nabla_{\mathbf{w}} J(\mathbf{w})$ points in which direction?

- ☐ The direction of steepest descent
- ☐ A direction that always points toward the global minimum
- ☒ The direction of steepest ascent
- ☐ A direction parallel to the contour lines



Question 12

1 / 1 pts

If the learning rate is too large, what is a likely consequence?

- ☐ The cost must decrease at every iteration
- ☐ The algorithm becomes the closed-form solution
- ☒ Updates may overshoot the minimum and move farther away
- ☐ The gradient becomes perpendicular to the contour lines for the first time



Question 13

1 / 1 pts

What is the mathematical definition of a feature mapping in the context of linear regression?

- ☒ A function that transforms each input vector into a new representation.
- ☐ A procedure that maps input features directly to their corresponding training targets.
- ☐ A strategy for reducing the total number of data points to limit computation time.
- ☐ An algorithm that selects the most important original features from a dataset.



Question 14

1 / 1 pts

If the λ in $\mathcal{E}(\mathbf{w}) + \lambda \mathcal{R}(\mathbf{w})$ is too small, then

- ☐ The regularizer dominates and the model may underfit
- ☒ The MSE term dominates and overfitting may remain
- ☐ The bias term is forced to zero
- ☐ The polynomial degree decreases automatically



Question 15

1 / 1 pts

Regularization is a modification of training that

- ☒ Chooses models using criteria beyond training error
- ☐ Forces all weights to have similar scale during training
- ☐ Guarantees zero training error on the dataset
- ☐ Removes the need to tune hyperparameters

Quiz Score: 15 out of 15